Wenbo Zhang

London, UK | ucacwzh@ucl.ac.uk | ftzhangwenbo@mail.scut.edu.cn | Google Scholar

Research Profile

I am a Ph.D. candidate working on **mobile and wearable sensing**, **multimodal sensor learning**, and **mobile health systems**. My research develops wearable systems integrating IMU, surface EMG, plantar pressure, and PPG to infer human motion, muscle effort, and physical interaction in real-world settings. I am particularly interested in robust, interpretable, and on-device learning, together with real-time feedback for health, fitness, and sport.

Research Experience

Visiting Ph.D. Researcher, University College London (UCL), London, UK *Mar 2026 – Present*
Supervisor: Prof. Jagmohan Chauhan

- Investigate multimodal wearable sensing and machine learning for human motion and muscle effort, with real-time feedback for health and exercise applications.
- Explore multimodal sensor learning, wearable agents, and efficient mobile and on-device deployment for context-aware and actionable feedback.

Ph.D. Researcher, South China University of Technology, Guangzhou, China *Sep 2023 – Present*
Supervisor: Prof. Zhanpeng Jin

- Develop wearable sensing systems integrating IMU, surface EMG, plantar pressure, and PPG, with optical motion capture used as ground truth for human motion and musculoskeletal analysis.
- Lead end-to-end studies spanning human-subject data collection, sensor synchronization, multimodal modeling, mobile deployment, and system evaluation.

Education

South China University of Technology, Guangzhou, China *Sep 2023 – Jun 2027*
Ph.D. Candidate in Information and Communication Engineering (Expected Jun 2027)

Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China *May 2021 – Jun 2023*
Joint M.S. Research Program in Computer Science

Hebei University, Baoding, China *Sep 2020 – Jun 2023*
M.S. in Computer Science

Hebei University of Science and Technology, Shijiazhuang, China *Sep 2016 – Jun 2020*
B.S. in Internet of Things Engineering

Selected Peer-Reviewed Publications

[P1] Wenbo Zhang, Chenxu Zhang, Yang Gao, and Zhanpeng Jin. "KineticsSense: A Multimodal Wearable Sensor Framework for Modeling Lower-Limb Motion Kinetics." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 9(3), Article 151, 2025. DOI

[P2] Lingde Hu†, **Wenbo Zhang**†, Wenkang Zhang, Yu He, Seokmin Choi, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. "PPGSpeech: A Wearable Silent Speech Interface Leveraging Neck-Worn Photoplethysmography." *IEEE Internet of Things Journal*, 13(4), pp. 6692–6703, 2026. DOI † Equal contribution.

[P3] Yang Gao, **Wenbo Zhang**, Junbin Ren, Ruihao Zheng, Yingcheng Jin, Di Wu, Lin Shu, Xiangmin Xu, and Zhanpeng Jin. "PressInPose: Integrating Pressure and Inertial Sensors for Full-Body Pose Estimation in Activities." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 8(4), Article 197, 2024. DOI

[P4] Junbin Ren, Ruihao Zheng, **Wenbo Zhang**, Dong She, Yuting Bai, Zhanpeng Jin, and Yang Gao. "Motion2Press: Cross-Modal Learning from IMU to Plantar Pressure for Gait Analysis." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 9(3), Article 124, 2025. DOI

Manuscripts in Revision and Under Review

[M1] Wenbo Zhang, Chenxu Zhang, Xingying Yan, Guanyu Xin, Yu He, Wenkang Zhang, Jagmohan Chauhan, Yang Gao, and Zhanpeng Jin. "MuscleSense: Making Muscle Effort Audible for Eyes-Free In-the-Loop Regulation." Revised manuscript under review at *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 2026.

[M2] **Wenbo Zhang**, Xingying Yan, Chenxu Zhang, Dong Huitian, Haoyang Li, Chenxu Zhu, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. “HALO: Inferring Hidden Activation–Loading Organization from Sparse Wearables.” Manuscript under review at *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 2026.

[M3] Lingde Hu[†], **Wenbo Zhang**[†], Seokmin Choi, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. “WatchForce: Wearables Can Tell How Strong You Grasp From Your Wrist.” Manuscript under revision following a major revision decision at *Computer Methods and Programs in Biomedicine*, 2026.

Additional Research Outputs

[A1] He Yu, Wenkang Zhang, **Wenbo Zhang**, Lingde Hu, Muhammad Adil, Yincheng Jin, and Zhanpeng Jin. “A Survey on Brain Decoding: Reconstruction of Audio-Visual Perception.” *ACM Computing Surveys*, in press, 2025.

[A2] Zhenchao Cui, **Wenbo Zhang**, Zhaoxin Li, and Zhaoqi Wang. “Spatial–Temporal Transformer for End-to-End Sign Language Recognition.” *Complex & Intelligent Systems*, 9(4), pp. 4645–4656, 2023. DOI

[PT1] **Wenbo Zhang** et al. “A Full-Body Pose Estimation Method and System Based on Pressure and Inertial Sensors.” Granted Chinese invention patent.

Core Research Competencies

- **Research Formulation and Scientific Strategy:** Formulating research questions in wearable sensing and mobile intelligence, and translating them into experimental designs and technical roadmaps.
- **Multimodal Wearable Intelligence:** Developing multimodal and cross-modal learning methods to infer human motion, muscle activity, force, and biomechanical states from sparse and heterogeneous wearable signals.
- **End-to-End Systems and Human-Centered Evaluation:** Building complete wearable sensing pipelines spanning sensor integration, synchronization, lightweight inference, smartphone deployment, and real-time feedback, together with human-subject evaluation and junior researcher mentoring.

Selected Research Projects and Proposals

Multimodal Biomechanics Representation Learning for Human Motion *2025 – Present*
Provincial Natural Science Foundation (General Program), funded; PI: Prof. Zhanpeng Jin. Contributed to the technical design of multimodal wearable sensing and interpretable biomechanical state modeling.

Human Motion Representation with Semantic–Physical Consistency *2026*
General Program proposal to the National Natural Science Foundation of China; PI: Prof. Zhanpeng Jin. Contributed substantially to the development of the overall research framework, formulation of key scientific questions, and design of the technical roadmap.

Digital Twin-Based Health Monitoring and Management for Older Adults *2025 – Present*
National Key R&D Program proposal under review. Contributed to the technical framework for unobtrusive sensing, physical-function assessment, musculoskeletal modeling, and personalized health management.

Teaching and Mentoring

Teaching Assistant, Introduction to Engineering *2024 – 2026*
- Supported project-based teaching and engineering design activities for first-year undergraduate students.
- Mentored two cohorts of first-year student teams in consecutive years; both cohorts received Second Prize in the annual course design competition.

Academic Service

- Reviewer for ACM IMWUT/UbiComp and AAAI.
- Contributed to the preparation and organization of *WearAgent 2026: The 1st International Workshop on Interactive AI for Personal Wearable Agents*, ACM UbiComp/ISWC 2026.
- Invited panelist, “Impact, Risks, and Security/Privacy in Ubiquitous Computing Research,” ACM UbiComp/ISWC 2025.
- Student Volunteer, ACM UbiComp/ISWC 2025, Helsinki, Finland.

Selected Honors

- China Scholarship Council Joint Ph.D. Scholarship, 2025.
- Silver Medal, Huawei Ascend AI Innovation Competition, 2023.
- Research Excellence Award for Graduate Students, 2023.
- Outstanding Graduate, 2023.